Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/22/2021
PI Name:	Allaway, Heather Ph.D.		
Project Title:	Will Use of Long-Acting, Reversible Contraceptives Mitigate Bone Loss in Female Rats During Simulated Exploration Class Missions?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70803-0001	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 TRI-RFA-17-01: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	01/01/2018	**End Date:**	12/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2020 per March 2020 report from TRISH (Ed., 4/7/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bloomfield, Susan Ph.D. (MENTOR: Texas A&M Research Foundation)		
Grant/Contract No.:	NNX16AO69A-P0204		
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Task Description:	POSTDOCTORAL FELLOWSHIP Astronauts who spend a long time in weightlessness typically lose bone and muscle mass -- which may be very dangerous for astronauts arriving on Mars. Hormonal contraceptive (HC) methods routinely used female astronauts, and women worldwide, to suppress ovarian function and menstrual cycling. Use of HC decreases natural estrogen and progesterone exposure and the positive influence of both hormones on bone health. There is a critical gap in knowledge surrounding long-acting, reversible contraceptive (LARC; progestin-only) use, an HC option chosen with increasing frequency, and bone geometry and cellular activity. There is no knowledge about the interaction of LARC use and microgravity on bone health. Specific Aim 1: Determine the dose dependent impact of an etonogestrel (ENG) LARC implant on ovarian function in intact, young adult female rats over 28 days of exposure. Specific Aim 2: Determine the combined effects of an ENG LARC implant and skeletal disuse (hindlimb unloading; HU) on bone health in intact, young adult, female rats.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will have impact with the shift from short to long duration and exploration class missions and has the potential to shift clinical practice. In female astronauts of reproductive age, the impact of spaceflight on gonadal function is complicated by the use of hormonal contraception to suppress the menstrual cycle pre-flight and during flight. Ground based studies have failed to directly account for the impact of long term use of hormonal contraception on bone health. We want to facilitate the improvement of best practice by working to ensure that long term use of hormonal contraception methods does not impair health. This study will be the first to explore the impact of LARC use on bone health using an animal model of microgravity exposure and will use measures of bone geometry, strength, and cellular activity. The results of this study will be translatable to the astronaut corps as well as millions of women around the world who use hormonal contraception. Additionally, this project is the development of a novel model to assess the impact of hormonal contraceptive use on bone health and future fracture risk.
Task Progress:	Astronauts who spend a long time in weightlessness typically lose bone and muscle mass - which may be very dangerous for astronauts arriving on Mars. Hormonal contraceptive (HC) methods are routinely used female astronauts, and women worldwide, to suppress ovarian function and menstrual cycling. Use of HC decrease natural estrogen and progesterone exposure and the positive influence of both hormones on bone health. There is a critical gap in knowledge surrounding long-acting, reversible contraceptive (LARC; progestin-only) use, an HC option chosen with increasing frequency, and bone geometry and cellular activity. There is no knowledge about the interaction of LARC use and microgravity on bone health. Specific Aim 1: Determine the dose dependent impact of an etonogestrel (ENG) LARC implant on ovarian function in intact, young adult female rats over 28 days of exposure. Animals arrived at 16 weeks of age, had the ENG LARC/placebo implant inserted at 20 weeks of age and were sacrificed at 24 weeks of age. The ENG LARC implants released doses of ENG at rates of 0.00 ug/d (placebo), 0.09 ug/d (low), 0.17 ug/d (medium), or 0.30 ug/d (high). Animal body weight and food consumption did not differ among treatment groups ($p>0.150$). H1A/B - Uterine and ovarian weights at sacrifice (a biomarker of estradiol exposure) did not differ among treatment groups ($p>0.150$). Serum estradiol was reduced in a treatment group study day interaction such that 7 days after ENG pellet insertion, the placebo group had higher estradiol compared to the low, medium, and high dose groups ($p=0.016$). H1C - Ovarian volume and number of ovulatory sized follicles (>400um) did not differ among treatment groups ($p>0.071$) but varied by study day ($p<0.050$). The ultrasound biomicroscope images were not clear in all animals to monitor the smallest follicle sizes we intended to assess. The number and area of new corpora lutea differed by study day ($p<0.050$). Evaluated as total number of days or percent total time, the time spent in proestrus, estrus, metestrus, and diestrus did not differ among the treatment groups prior to implant insertion ($p>0.30$). Following implant insertion, time spent in estrus was reduced and diestrus increased for the medium and high dose treatment groups ($p<0.001$), with no differences in time spent in proestrus or metestrus ($p>0.08$). In the first 2 weeks following implant insertion there is an apparent dose dependent shift toward irregularity and acyclicity ($p<0.001$), which continued into the last 2 weeks of implant insertion ($p=0.02$). The shift in cycle classifications was similar to the shifts observed in women over the 3 years of implant use. H1D (exploratory) - P1NP varied with time but not by treatment group and was lower 4 weeks after pellet implantation compared with the week before implantation. CTx did not vary significantly with time or by treatment group. Specific Aim 2: Determine the combined effects of an ENG LARC implant and skeletal disuse (hindlimb unloading; HU) on bone health in intact, young adult, female rats. Based on the more controlled suppression of estrus cyclicity, the high dose (0.30 ug/d) of ENG LARC was chosen as the dose to be used in Specific Aim 2. Animals arrived at 16-18 weeks of age, had the ENG LARC/placebo implant inserted at 20-22 weeks of age. The weight bearing/HU phase of the study was initiated at 21-23 weeks of age and sacrifice occurred at 27-29 weeks of age. H2A: LARC implantation did not alter the expected HU induced changes in volumetric bone mineral density (BMD) or cortical thickness at the metaphysis or diaphysis of the tibia. Tibia mechanical strength was partially protected by LARC implantation ($p=0.012$). H2B: blunt the decreases in bone formation rate and osteoid surface and blunt the increases in osteoclast surface observed with 6 weeks of HU; At the tibia diaphysis HU animals had a reduced cortical bone formation rate per unit of bone surface (BFR/BS) on the periosteal surface ($p=0.015$) but not the endosteal surface ($p=0.103$). At the tibia metaphysis, cancellous BFR/BS did not differ among treatment groups ($p=0.741$). At the tibia metaphysis, there were no differences in osteoid surface per bone surface (OS/BS; $p=0.833$) or osteoclast surface (OcS/BS; $p=0.267$) among treatment groups. Based on 2D histomorphometric analysis, bone volume/tissue volume (BV/TV) was suppressed in the PL-HU animals compared with the PL-WB animals ($p=0.023$).

	H2C (exploratory) - Immunohistochemical assays for estrogen and progesterone receptor presence in osteocytes has been completed and microscopy analysis underway. Specific Aim 3: Determine the combined effects of LARC and overloading on the health of the humeri of female rats in HU. H3A: Micro-Computed Tomography (uCT) of the humeri is underway (delayed due to COVID-19 enforced lab closures) and when completed mechanical testing will occur. H3B: Bone formation rate was reduced on the endocortical surface of the humerus diaphysis for HU animals (p=0.008) and LARC animals (p=0.047). Bone formation rate was not impacted by LARC use or HU on the periosteal surface of the diaphysis or in the cancellous space of the metaphysis of the humerus. Humeri were processed for static histology and analyses are awaiting time on the microscope. Specific Aim 4: Determine the combined effect of a LARC implant and HU on the gastrocnemius muscle of female rats. We were not able to start this specific aim due to delays caused by lab moves and COVID-19 lab closures. Key findings did not lead to a revision of the original aims or hypotheses. Costs, availability of assays, lab moves, and the global pandemic has delayed completion or prevented the completion of some portions of the project. The paper for Specific Aim 1 has been submitted for publication. The funding for this project is complete; however, we will continue working to finish Specific Aims 2 and 3 within the next 4 months. We are unsure when or if we will be able to complete Specific Aim 4.
Bibliography Type:	Description: (Last Updated: 01/28/2026)
Abstracts for Journals and Proceedings	Allaway HCM, Elizaondo JP, Little SE, Hogan HA, Bloomfield SA. "LARC implant use does not exacerbate bone loss associated with hindlimb unloading." 67th Annual Meeting of the American College of Sports Medicine, Virtual, May 26-30, 2020. Medicine & Science in Sports and Exercise. 2020 Jul;52(7S):73-4. https://doi.org/10.1249/01.mss.0000670832.15037.31 , Jul-2020
Abstracts for Journals and Proceedings	Martone M, Irving B, Spielmann G, Allaway H. "Effects of hindlimb unloading and long acting reversible contraception on markers of mitophagy in muscle: 2310." American College of Sports Medicine 2022 Annual Meeting, San Diego, California, May 31-June 4, 2022. Medicine & Science in Sports & Exercise. 2022 Sep;54(9S):675-6. https://doi.org/10.1249/01.mss.0000883508.54288.73 , Sep-2022
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Articles in Peer-reviewed Journals	Allaway HCM, Pierson RA, Invik J, Bloomfield SA. "A rodent model of human dose-equivalent progestin-only implantable contraception." Reprod Biol Endocrinol. 2021 Mar;19(1):47. https://doi.org/10.1186/s12958-021-00729-w ; PMID: 33752672; PMCID: PMC7983206 , Mar-2021